

Supply Chain/Logistics Domain Cheatsheet for BA

1. Domain Overview

The Supply Chain & Logistics domain manages the **flow of goods, information, and finances** from raw material sourcing to final delivery. It covers procurement, manufacturing, warehousing, transportation, and distribution. Business analysts work on ERP/SCM implementations, demand forecasting, inventory optimization, warehouse automation, and logistics visibility platforms.

2. Key Stakeholders

Stakeholder	Role / Interest
Suppliers & Vendors	Provide raw materials, goods, services
Manufacturers	Convert raw materials into finished goods
Logistics Providers (3PL/4PL)	Handle warehousing, transportation, last-mile delivery
Distributors & Wholesalers	Move goods across regions/channels
Retailers & E-commerce	Sell products to end customers
Customers (B2B/B2C)	Expect timely, accurate deliveries
Regulators & Customs	Enforce trade laws, tariffs, compliance
Finance Teams	Manage procurement spend, freight costs
IT/SCM Vendors	Provide ERP, WMS, TMS, SCM, and analytics tools

3. Common Processes

- **Procurement & Sourcing:** Vendor selection, purchase orders, supplier contracts.
- **Production Planning & Scheduling:** Forecasting demand, managing production runs.
- **Inventory Management:** Stock control, reorder levels, safety stock optimization.

- **Warehousing & Fulfillment:** Receiving, picking, packing, storage, distribution.
- **Transportation Management:** Route planning, carrier selection, shipment tracking.
- **Last-Mile Delivery:** Ensuring on-time delivery to end customers.
- **Reverse Logistics:** Returns, recycling, and recalls.
- **Customs & Trade Compliance:** Duties, tariffs, cross-border clearance.
- **Supply Chain Finance:** Vendor payments, freight settlements.

v4. Key Systems/Applications

- **ERP Systems:** SAP, Oracle, Microsoft Dynamics (end-to-end supply chain management).
- **WMS (Warehouse Management Systems):** Manage storage, picking, inventory accuracy.
- **TMS (Transportation Management Systems):** Optimize routing, carrier management.
- **SCM Platforms:** End-to-end visibility (Kinaxis, Blue Yonder, Manhattan Associates).
- **Order Management Systems (OMS):** Handle multi-channel orders.
- **Supplier Portals:** Vendor collaboration, procurement automation.
- **IoT/Telematics:** Real-time tracking of goods & vehicles.
- **Analytics & Demand Forecasting Tools:** AI/ML-driven predictions.

5. Regulatory/Compliance Factors

- **Customs & Trade Laws:** Import/export duties, tariffs, documentation.
- **Incoterms:** Standardized international shipping terms.
- **Environmental Regulations:** Emission standards, green logistics compliance.
- **Occupational Safety:** OSHA, warehouse safety protocols.
- **Data Privacy:** GDPR/CCPA in customer and supplier data.
- **Food/Pharma Supply Chains:** FDA/EMA compliance for cold chain integrity.

6. Common Pain Points

- Supply chain disruptions (pandemics, geopolitical issues).
- Forecasting inaccuracies leading to stockouts/overstocking.
- High logistics costs, especially in last-mile delivery.
- Lack of real-time visibility across suppliers & carriers.
- Customs delays and regulatory complexity.
- Legacy ERP/WMS systems with poor integration.
- Rising sustainability and carbon compliance requirements.

7. Opportunities for BAs

- Map & redesign end-to-end procurement-to-delivery workflows.
- Implement real-time visibility platforms for shipments.
- Automate demand forecasting & inventory replenishment.
- Optimize last-mile delivery with route optimization tools.
- Enable supplier collaboration portals for procurement.
- Define KPIs for cost-to-serve, order fill rate, and on-time delivery.
- Support sustainability reporting & green logistics initiatives.

8. Commonly Used Terminology (Quick Glossary)

- **3PL/4PL:** Third/Fourth Party Logistics providers.
- **Lead Time:** Time from order placement to delivery.
- **Fill Rate:** % of customer demand met without stockouts.
- **Safety Stock:** Extra inventory kept to prevent shortages.
- **SKU:** Stock Keeping Unit, unique identifier for a product.
- **Incoterms:** International commercial terms defining buyer/seller responsibilities.
- **Reverse Logistics:** Handling returns, recycling, disposal.
- **Cold Chain:** Temperature-controlled supply chain for perishable goods.

9. Emerging Trends

- **AI/ML in Demand Forecasting:** Improved accuracy in planning.
- **Blockchain for Supply Chain Transparency:** Tamper-proof traceability.
- **IoT & Telematics:** Real-time asset and fleet monitoring.
- **Green Logistics:** Electric fleets, carbon-neutral delivery models.
- **Digital Twins:** Virtual replicas of supply chains for simulation.
- **Hyperautomation in Warehousing:** Robotics, drones, automated picking systems.
- **Resilient Supply Chains:** Diversification of suppliers, nearshoring strategies.

